

## Quiz 03

**Instructions:** Answer the questions with precise definitions. Your definitions should either be stated exactly as in the lecture or textbook, or be clearly equivalent to them. Having the “right idea” but not a precise formulation will not receive credit.

**Problem 1.** Let  $Y_1, Y_2$  be random variables. What does it mean for  $Y_1$  and  $Y_2$  to be **jointly continuous**?

**Problem 2.** Define the **expected value** of a continuous random variable  $Y$ . Make sure to state any conditions that must be true for the expected value to exist.